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一、專題名稱

Taiwan – Global Footprint Dashboard

二、專題介紹

This project delivers an interactive **R Shiny** application that allows users to explore how Taiwan's economy, democracy, tourism and labour market interact with—and influence—the wider world. Six tabs (Macro, Democracy, Stock, Foreigners, Salary, Data) share a common sidebar so users can slice the same time window and peer countries throughout the app.

三、專題製作動機

1. **Economic curiosity** — As a student in international economics, I kept seeing headlines about Taiwan's semiconductor dominance; I wanted hard numbers.
2. **Tech fascination** — Companies I follow (NVIDIA, Apple) depend on TSMC wafers. Quantifying that linkage felt like a good cap-stone for our R class.
3. **Civic angle** — With democracy scores diverging between Taipei and Beijing, visualising the gap gives context to current events.

四、專題操作說明

1. Prerequisites

```
if (!require(tidyverse)) install.packages("tidyverse")
if (!require(shiny))     install.packages("shiny")
```

1. Start the app

```
shiny::runApp("app.R")
```

1. Sidebar controls

2. *Year range* slider drives every plot.
3. *Compare GDP / Democracy with:* choose peer economies.
4. Tab-specific inputs (e.g., stock tickers, salary variables) appear when needed.

5. **Downloads**—Two buttons export the current Democracy slice and Salary slice as CSV for further analysis.

五、專題程式流程

Below is a condensed pseudocode of the server logic, followed by a Mermaid flowchart.

```
# --- DATA CLEANING -----
read_clean_csv <- function(path, ...) {
  # Helper to clean column names and suppress warnings
  readr::read_csv(path, show_col_types = FALSE, ...) %>%
    janitor::clean_names()
}

# GDP Taiwan -----
gdp_tw <- read_clean_csv("data/GDP Taiwan 2001 - 2020.csv") %>%
  rename(year_raw = 1) %>%
  mutate(year = parse_number(year_raw),
         gdp_usd_bn = gdp_nt_mn / 30 / 1e3) %>%
  select(year, gdp_usd_bn)

# Reactive slice example -----
gdp_slice <- reactive({
  bind_rows(
    gdp_tw %>% filter(year %in% input$yrange[1]:input$yrange[2]) %>%
      mutate(country = "Taiwan"),
    gdp_world_long %>%
      filter(country %in% input$country_compare,
             year %in% input$yrange[1]:input$yrange[2])
  )
})
```

```
flowchart LR
  A[CSV on Kaggle] -->|read_clean_csv| B(Clean tibble)
  B --> C[Reactive slices] --> D[Plots & Tables]
  D --> E[CSV Downloads]
```

六、結論

Findings

- Taiwan's GDP doubled from 2006–2023, while its CPI stayed tame—evidence of balanced growth.
- Democracy Index shows Taiwan climbing into the *Full Democracy* tier as China slides further down.

- NVIDIA's 2023 boom created parallel lifts in TSMC, Apple and Microsoft—demonstrating semiconductor coupling.
- Visitor analysis reveals that Mainland tourists remain the largest bloc, so political friction carries direct tourism risk.
- Salary data indicate Taiwanese tech talent works longer hours than U.S. peers yet earns less—fuel for potential brain-drain.

Reflection Building the dashboard was a triathlon of data cleaning, reactive programming and storytelling. The trickiest bug came from an `adjclose` typo inside the NVIDIA CSV; my `clean_stock()` function now falls back to a generic column match. From this I learned to **write defensive functions** and sprinkle assertions early. If I had more time, I would add forecasting modules (e.g., ARIMA on GDP) and Dockerise the deployment for reproducibility.

七、附錄

- **Data source:** All raw datasets were downloaded from open Kaggle competitions or public kernels.
- **Key packages:** `tidyverse`, `shiny`, `janitor`, `lubridate`, `scales`, `ggplot2`, `mermaid`.
- **Variable notes:**
 - `gdp_usd_bn` — GDP in *billions* of constant US dollars.
 - `demo_score` — Economist Intelligence Unit index (0 = worst, 10 = best).
 - `adj_close` — Split- and dividend-adjusted stock price.
 - `salary` — Binary: `<=50K`, `>50K` annual income bracket.